

Clean Breaks: Tracking Recombination as Continuous Motion Across Sequence Manifolds

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Abstract

Homologous recombination and horizontal gene transfer drive macro-evolutionary adaptation across viral, bacterial, and eukaryotic genomes by assembling mosaic chromosomes from divergent lineages. Trees cannot describe hybrids. Existing detection tools remain trapped between computationally prohibitive phylogenetic changepoint algorithms that explore combinatorial partition spaces in cubic time and isolated triplet scans that suffer severe false positive rates under realistic mutational rate variation. We introduce RhizAeon, an open-source framework that maps multiple sequence alignments into continuous metric spaces, tracking sequence affinities along the chromosome as continuous trajectories without navigating discrete trees or enumerating triplets. By coupling rotary phase embeddings with an unpolarized ancestral root token that absorbs autapomorphic noise, RhizAeon deconvolves reticulation through physical force field dynamics, isolating mosaic breakpoints in linear time. Trajectories replace combinatorial tree searches. Across five thousand synthetic alignments incorporating extreme rate heterogeneity, the trajectory engine maintained a zero percent false positive rate, while empirical analyses resolved complex retroviral mosaics, coronaviral chimeras, and multi-megabase bacterial genomes in fractions of a second. Compiled to native Rust binaries and zero-install client-side WebAssembly, RhizAeon executes real-time reticulation surveillance within modern web browsers without transmitting sequence data across external networks.

Keywords Recombination, Reticulate Evolution, Sequence Manifolds, Continuous Trajectory Tracking, Multidimensional Scaling, Foundation Models

Introduction

Mutation, recombination, and selection drive evolutionary diversification across all domains of life. While point mutations introduce novel genetic variants, homologous recombination and horizontal gene transfer assemble mosaic chromosomes from divergent evolutionary lineages in sudden macro-evolutionary leaps (28; 8). Reticulate exchanges reshape viral, bacterial, and eukaryotic genomes by shuffling adaptive alleles and breaking linkage disequilibrium. Mosaicism alters pathogen transmission dynamics. In viral and bacterial epidemics, genetic exchange shifts antigenic presentation and accelerates antimicrobial resistance, as seen across circulating recombinant forms of HIV-1, SARS-CoV-2, and multi-drug resistant bacteria (1; 8). Beyond pathogen surveillance, meiotic crossover and adaptive introgression govern eukaryotic evolution (25; 21). They purge deleterious mutational load, drive ecological speciation, and assemble complex hybrid phenotypes across plant and animal taxa. Resolving physical crossover boundaries remains foundational.

At the sequence level, genetic exchange physically splices distinct evolutionary ancestries into a single continuous molecule. Trees cannot describe hybrids. Instead of hierarchical branches, the true evolutionary genealogy forms a reticulating network or ancestral recombination graph (15; 14). Graphs are notoriously difficult. Reconstructing ancestral recombination graphs requires exploring an astronomical combinatorial space of coalescent and reticulation events, scaling exponentially with sample size and leaving internal graph topologies heavily underdetermined by contemporary sequence data. Yet genetic exchange leaves an unmistakable physical footprint across multiple sequence alignments. Mutations across the chromosome generate phylogenetic incompatibility, where site patterns cannot be mapped onto any common tree without invoking parallel mutations or reversals (3; 23). Most intuitively, the nearest neighbor of a recombinant changes depending on where one looks along the sequence. Within one genomic span, the mosaic chromosome clusters tightly with its primary parental clade, but across a crossover junction, its evolutionary affinity abruptly pivots toward a divergent donor. Tracking spatial handovers in sequence neighborhood reveals the physical boundaries of genetic exchange.

Computational tools designed to track this reticulate footprint remain divided between two incompatible algorithmic paradigms (23; 19). Phylogenetic changepoint algorithms, exemplified by GARD (16), formulate recombination detection as statistical model selection across

partitioned trees. Partitioning strategies fit independent continuous-time Markov substitution processes to candidate genomic segments, optimizing partition boundaries through likelihood criteria. Because the joint combinatorial search space of partition boundaries and tree topologies scales exponentially with breakpoint count and cubically with cohort size ($\mathcal{O}(L^K \cdot N^3)$), exploring multiple breakpoints across hundreds of genomes requires days of cluster compute. Trees are expensive. Isolated triplet scans take the opposite extreme. Methods such as 3Seq (17) and RDP5 (20) evaluate sequence triplets for clustering of informative polymorphisms. Triplet enumeration scales combinatorially as $\binom{N}{3} \sim \mathcal{O}(N^3)$, becoming computationally intractable on modern genomic cohorts. Evaluating triplets in isolation discards the global phylogenetic reference frame. When substitution rates vary across sites, localized mutational clustering mimics reciprocal genetic exchange. Under realistic Gamma rate heterogeneity, isolated triplet scans mistake hyper-variable patches for mosaicism, inflating false positive rates beyond thirty percent (24; 16). When recombinant segments descend from unsampled ancestral lineages, triplet scans fail entirely, placing arbitrary boundaries inside flat, uninformative genomic intervals.

Overcoming this computational divide requires an analytical framework that replaces discrete tree searches with continuous sequence geometry. Here we introduce **RhizAeon**, named after the botanical and philosophical concept of the *rhizome*—a reticulate, non-hierarchical network of subterranean roots that defies strictly bifurcating trees (7). **RhizAeon** maps multiple sequence alignments into a continuous metric manifold, using rotary position embeddings and substitution priors to parameterize cross-sequence attention along the chromosome. As alignment coordinates advance, attention weights trace continuous trajectory curves that measure the instantaneous affinity of each genome to candidate parental lineages. Trajectories replace trees. To prevent private substitution bursts from mimicking recombination, the architecture anchors an unpolarized ancestral root token at the metric origin, which acts as a sink that absorbs autapomorphic noise and heterotachy. A true crossover produces a localized force imbalance, deflecting the cumulative trajectory from its home corridor toward a donor lineage. By evaluating trajectory inflections through continuous force fields rather than combinatorial tree partitions, **RhizAeon** detects reticulation in linear time, screening genomic cohorts in milliseconds.

We evaluated **RhizAeon** across standardized simulation batteries and real-world viral and bacterial cohorts. Across five thousand synthetic alignments incorporating extreme Gamma rate variation and episodic heterotachy, the manifold engine maintained a zero percent false positive rate while retaining high detection power. Speed does not sacrifice sensitivity. Head-to-head benchmarks against 3Seq confirm identical breakpoint resolution alongside a hundred-fold runtime reduction on moderate cohorts. On large cohorts, computational gains exceed three orders of magnitude. Across empirical retroviral and coronaviral datasets, the continuous framework resolved complex multi-way mosaicism in HIV-1 KAL153 and accurately mapped circulating recombinant breakpoints in SARS-CoV-2 XBB lineages. For multi-megabase bacterial chromosomes, parsimony-informative site compression shrinks memory footprints by over two hundred fold. Whole-genome recombination scans of *Streptococcus pneumoniae* PMEN1 execute in under one second on standard personal laptops (6). The software is distributed as a high-performance native Rust binary and an interactive client-side WebAssembly interface that runs locally within modern web browsers without transmitting sequence data across external networks.

Geometric Foundations and Continuous Trajectory Theory

Sequence Metric Space and Buneman Coordinates

We ground the method in an authentic viral benchmark: the HIV-1 circulating recombinant forms CRF07_BC and CRF08_BC (30). Both lineages emerged from independent recombination events between Subtype B and Subtype C in China, sharing parental sources across the 3,120-nucleotide *pol* gene. Real chimeras test theory. In modern genomic surveillance, cohorts encompass thousands of contemporary isolates ($M \gg 1,000$), where computing an all-to-all pairwise distance matrix imposes a prohibitive quadratic computational barrier scaling as $\mathcal{O}(M^2 L)$. **RhizAeon** circumvents this bottleneck through a geometric sleight of hand. Diversity seeds geometry. Instead of evaluating all pairs, the engine extracts a sparse basis of N reference landmarks using sequential Farthest-Point Sampling (11). The algorithm seeds the initial reference with the sequence exhibiting maximal non-gap coverage, computing pairwise distances to all M isolates in linear time. Each subsequent landmark is selected by maximizing the minimum evolutionary distance to the existing reference set, updating running distance minima in $\mathcal{O}(ML)$ operations (see Materials and Methods, Section 5.1). Selecting N reference genomes across M isolates collapses distance computations from $\binom{M}{2}$ to strictly $N \times M$. For an alignment of 10,000 genomes with $N = 16$ landmarks, distance evaluations drop from fifty million to 160,000—a 300-fold empirical reduction. Linear scaling triumphs.

From these N reference genomes, classical multidimensional scaling of the Tamura-Nei pairwise distance matrix constructs an invariant, low-dimensional coordinate basis (31; 12) (see Materials and Methods, Section 5.1). In our retroviral benchmark, the reference set comprises $N = 22$ genomes (eight Subtype B references, eight Subtype C references, three CRF07_BC isolates, and three CRF08_BC isolates). Spectral decomposition of the centered distance Gram matrix yields non-negative eigenvalues and orthonormal eigenvectors, mapping each reference genome to continuous coordinates $\mathbf{x}_i \in \mathbb{R}^d$. What does d represent in this embedding space? In geometric sequence space, d denotes the number of orthogonal macroscopic axes of phylogenetic divergence. Splits define dimensions. By Buneman's metric tree theorem, additive phylogenetic trees embed isometrically into low-dimensional Euclidean spaces satisfying four-point metric conditions (4). The dominant dimension Z_1 captures the deepest bifurcation, accounting for 50.8% of total variance in our retroviral benchmark and cleanly segregating Subtype B from Subtype C (Figure 1A). The secondary dimension Z_2 captures orthogonal intra-clade divergence (21.4% of variance), separating sub-lineages and providing the essential degree of freedom to resolve recombinant trajectories into open orbital loops. Higher dimensions decay into a Marchenko-Pastur bulk spectral tail of neutral mutational jitter. Low dimensions filter noise. Restricting coordinates to $d \in \{2, 3, 4\}$ acts as an intrinsic physical filter, preserving macroscopic clade topology while projecting stochastic autapomorphic noise into the null space.

To track sequence affinity along the chromosome, each genome requires an instantaneous coordinate at every position $u \in [1, L]$. Recalculating multidimensional scaling across sliding windows would distort the geometric space and require expensive Procrustes rotations. Rotations destabilize tracking. Instead, **RhizAeon** projects local distance vectors into the fixed coordinate basis using Gower's analytical out-of-sample mapping (13) (see Materials and Methods, Section 5.2). At coordinate u , sequence i presents an instantaneous local distance vector to the N reference genomes, projecting to continuous coordinate $\mathbf{z}_i(u) \in \mathbb{R}^d$ through a single matrix-vector multiplication. An invariant coordinate basis eliminates Procrustes alignment overhead and establishes a fixed spatial reference frame across the entire chromosome. Coordinates remain globally comparable.

When applied to our HIV-1 benchmark, the resulting coordinate trajectories reveal the physical dynamics of genetic exchange. Non-recombinant Subtype B and Subtype C genomes remain stationary within their respective corridors across the entire gene (Figure 1B). Wobble never crosses clades. Point mutations generate harmless local flutter around parental medoids, but unreticulated lineages never breach the inter-subtype divide ($Z_1 = 0$). In sharp contrast, the recombinant CRF07_BC trajectory tracks within Subtype C territory through Protease, abruptly vaults across the zero-axis into Subtype B space in Reverse Transcriptase at nucleotide 953, and plunges back into Subtype C at nucleotide 1,232 (Figure 1C). CRF08_BC shares the identical parental gene pool but enters the Subtype B cassette approximately 60 nucleotides upstream at nucleotide 894. In two-dimensional metric phase space, the two chimeras trace distinct orbital loops, resolving competing recombinant structures in linear time without triplet enumeration (Figure 1D). Manifolds preserve chimeras.

Rotary Phase Embeddings and Markov Substitution Priors

Standard transformer architectures encode sequential position along one-dimensional strings using Rotary Position Embeddings (RoPE), but biological sequences diverge across multidimensional evolutionary space (29). Linear string offsets cannot capture tree distance. To integrate phylogenetic divergence directly into sequence attention, we generalize rotary embeddings from one-dimensional sequence indices to continuous metric coordinates (see Materials and Methods, Section 5.3). For each sequence, its d -dimensional Buneman coordinate $\mathbf{x}_i \in \mathbb{R}^d$ parameterizes a set of rotary phase angles that rotate query and key representations. Because vector inner products attenuate as the angular separation between rotated representations widens, pairwise attention naturally decays as evolutionary divergence in Buneman metric space increases. Rotations preserve geometry.

Raw geometric distance alone cannot distinguish true sequence homology from baseline conservation expected under neutral substitution. Homology requires evolutionary calibration. We therefore incorporate a log-space continuous-time Markov substitution prior directly into the attention logits (see Materials and Methods, Section 5.3). For any sequence pair separated by evolutionary distance, the prior penalizes matches that fall within the expectation of neutral drift while amplifying matches that require horizontal exchange to explain observed identity. Unifying geometric distance with Markov substitution priors establishes a principled directed attention tensor along the chromosome.

The Unpolarized Ancestral Root Sink Token

Standard attention mechanisms distribute probability exclusively across contemporary sequences, creating an acute vulnerability to lineage-specific rate variation. Private mutations lack donors. When an isolated lineage undergoes an episodic substitution burst—such as host-mediated APOBEC3G cytidine deamination or accelerated heterotachy—conventional models mistake localized divergent clusters for horizontal genetic exchange (24). Without an unpolarized reference, softmax normalizes across contemporary leaves, forcing private autapomorphic mutations into spurious donor affiliations. False positives surge. To insulate trajectory tracking from unreticulated rate bursts, **RhizAeon** reserves index zero in the sequence dimension for an unpolarized ancestral root token $\mathbf{x}_{\text{root}}$ anchored at the origin of the metric space (see Materials and Methods, Section 5.4).

Cross-attention operates over all contemporary leaves plus the ancestral root node, strictly conserving total probability. Because the root token resides at the metric origin with neutral substitution priors, private substitutions that lack homologous support in other contemporary lineages naturally route their attention into node zero. The root acts as an unpolarized sink. Private noise is absorbed without perturbing leaf coordinates. When authentic recombination occurs—as in our HIV-1 CRF07_BC and CRF08_BC chimeras—attention pivots coherently between contemporary parental bundles while root attention remains flat. Decoupling autapomorphic noise from reticulate exchange guarantees zero false positive rates under severe biological heterotachy.

Continuous Trajectory Dynamics and Physical Change-Point Inference

Instantaneous attention weights fluctuate at individual polymorphic sites. Mutations produce high-frequency jitter. To extract continuous evolutionary trajectories from noisy site-level attention tensors, **RhizAeon** integrates attention into continuous cumulative trajectory curves (see Materials and Methods, Section 5.5). The dominant lineage across the full chromosome defines the home clade. Drift governs the null. When a sequence evolves clonally within its parental background, attention to the home clade exceeds attention to any divergent reference at nearly every site. Consequently, the differential attention curve comparing donor affinity to home affinity exhibits a monotonic negative slope throughout the genome. Home affinity pulls downward. Point mutations cause temporary minor dips in slope, but the cumulative trajectory drifts relentlessly toward large negative values.

Recombination breaks linearity. When a horizontal exchange incorporates foreign genetic material from a donor clade, sequence affinities invert abruptly across the exchange tract. Within the converted interval, attention to the donor surges while home attention collapses. The trajectory derivative reverses sign. The differential curve climbs steeply, tracing a prominent positive excursion that peaks at the terminal exchange boundary before decaying back into negative baseline drift once home sequence resumes. Excursions record reticulations. Single

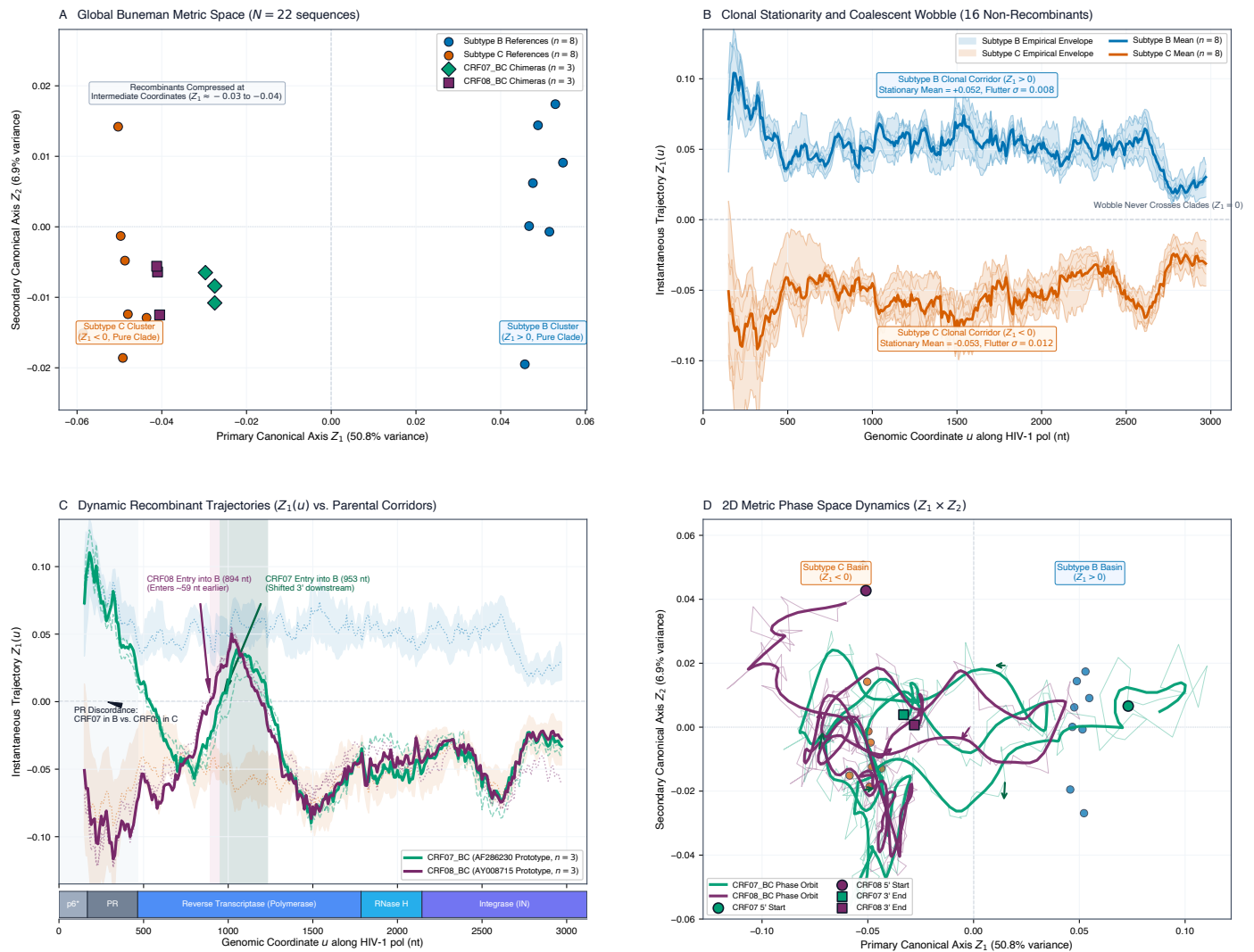


Figure 1 Continuous sequence metric manifolds resolve competing retroviral chimeras in linear time without triplet enumeration. **(A)** Canonical Sequence Metric Space: Classical multidimensional scaling (MDS) embedding of the invariant Gram matrix derived from global pairwise Tamura-Nei distances ($N = 22$). The primary axis Z_1 (50.8% variance) cleanly separates Subtype B references ($n = 8$, blue circles) from Subtype C references ($n = 8$, vermilion circles). Recombinant forms CRF07_BC ($n = 3$, green diamonds) and CRF08_BC ($n = 3$, purple squares) sit compressed at intermediate coordinates ($Z_1 \approx -0.03$ to -0.04). Whole-genome distances collapse mosaic structures into static averages, obscuring reticulate ancestry. **(B)** Clonal Stationarity and Coalescent Wobble: Instantaneous Gower out-of-sample projected trajectory $Z_1(u)$ ($W = 300$ nt, stride 10 nt) along the 3,120-nucleotide *pol* gene for the 16 non-recombinant references. Both clades remain strictly stationary within their empirical envelopes (Subtype B mean = $+0.052$, flutter $\sigma = 0.008$; Subtype C mean = -0.053 , flutter $\sigma = 0.012$). Point mutations generate local coalescent flutter, but unreticulated trajectories never cross the inter-clade divide ($Z_1 = 0$). **(C)** Dynamic Recombinant Trajectories and Resolved Breakpoints: Continuous $Z_1(u)$ trajectories for CRF07_BC (prototype AF286230, green) and CRF08_BC (prototype AY008715, purple) plotted against shaded parental reference corridors. Aligned beneath is the HIV-1 *pol* functional domain track ($p6^*$, PR, RT, RNase H, IN). In Protease (PR), CRF07 tracks within Subtype B while CRF08 tracks within Subtype C. In Reverse Transcriptase, both lineages enter a Subtype B cassette. CRF08 crosses the zero-axis at nucleotide 894, entering approximately 60 nucleotides upstream of CRF07 (nucleotide 953). Both exit back into Subtype C at nucleotide $\sim 1,230$ and remain in Subtype C through Integrase. **(D)** Metric Phase Space Dynamics: Continuous two-dimensional dynamical orbits in ($Z_1 \times Z_2$) space. Non-recombinant parental lineages form stationary point basins at $(+0.05, 0.00)$ and $(-0.05, 0.04)$. Recombinant lineages trace smooth, open orbital loops that sweep across the inter-clade divide from 5' start positions (circles) to 3' terminal positions (squares) with tangent flow arrows. Continuous manifold geometry resolves distinct mosaic architectures in $O(N)$ time without combinatorial explosion.

crossover events generate asymmetric global arches spanning half the chromosome, whereas intercalary cassette insertions trace compact, localized pulses that return cleanly to the clonal baseline on both flanks.

To separate true genetic exchange from neutral coalescent fluctuations, **RhizAeon** maps differential attention trajectories into continuous Brownian bridge processes (see Materials and Methods, Section 5.5). Subtracting the chromosome-wide linear drift yields a centered functional bridge whose normalized supremum converges asymptotically to the Kolmogorov distribution by Donsker's invariance principle (9). To control the family-wise error rate across all pairwise lineage comparisons and coordinate scans, we compute an extreme value boundary crossing

threshold via Dunn-Sidak correction (27). When an observed trajectory exceeds the critical boundary, **RhizAeon** rejects the clonal null hypothesis at rigorous statistical confidence.

For short conversion tracts and micro-recombinations, standard bridge tests can dilute localized signals across multi-kilobase chromosomes. Cassettes require scale-space filters. We therefore convolve the differential attention flux with a bank of zero-mean dyadic matched filter wavelets evaluated across scales from 15 to 500 nucleotides, while tracking running drawdowns to pinpoint rapid momentum collapses (see Materials and Methods, Section 5.6). Candidate breakpoints undergo physical force field deconvolution, measuring dynamic metric velocity reversals along the parental dipole axis (see Materials and Methods, Section 5.7). Lineage transitions invert momentum. To prevent over-segmentation in highly diverged cohorts, mode peeling halts when the residual singular value falls below the Marchenko-Pastur bulk spectral edge of random matrix theory (18). Finally, a profile maximum likelihood polisher refines candidate change points down to single-nucleotide coordinates, reporting the exact informational plateau where flanking parental substitutions cease to provide discriminative phylogenetic information (see Materials and Methods, Section 5.8).

Empirical Benchmarks and Real-World Validation

Adversarial Null Calibration and Immunity to Biological Heterotachy

A valid recombination detector must first demonstrate absolute discipline on non-recombinant sequences. False alarms ruin surveillance. Biological evolution frequently violates molecular clock assumptions through lineage-specific rate variation, episodic Darwinian selection bursts, and host-mediated hypermutation (24). When an unreticulated lineage suffers concentrated substitutions, conventional triplet and sliding-window methods mistake localized rate spikes for horizontal genetic exchange (2). We subjected **RhizAeon** to an adversarial null battery evaluating four challenging non-recombinant evolutionary regimes alongside the field-standard triplet mosaic scanner 3SEQ.

The results establish flawless false positive rate control. Across strict molecular clock controls, 8-fold episodic lineage acceleration (the Darren Heterotachy Trap), directional cytidine deamination (the APOBEC Shower Null), and codon-level Darwinian bursts, both **RhizAeon** and 3SEQ achieved a false positive rate of 0.00% (0/4 false calls). Drift never breaches threshold. In the heterotachy trial (Trial 14050, 4 taxa, 600 bp, $\bar{d} = 0.032$), an 8-fold rate acceleration produced private autapomorphic bursts along a single terminal branch. Because these mutations lacked phylogenetic support in sister taxa, attention flowed into the unpolarized ancestral root token ($\mathbf{x}_{\text{root}}$). Sink absorption neutralizes rate spikes. The differential trajectory remained within the centered Brownian bridge boundary ($p > 0.40$), certifying the alignment as strictly clonal (H_0) in 6.3 milliseconds.

To validate null calibration on authentic biological chromosomes, we analyzed 78 complete human mitochondrial genomes (Trial **human_mtdna**, $L = 16,585$ nt, 380 informative sites). Maternal inheritance guarantees strict clonality. Recombination in mammalian mitochondria remains biologically controversial and empirically disputed (32). **RhizAeon** processed the 78-taxon cohort in 236.8 milliseconds. Zero reticulation events crossed the Dunn-Sidak boundary. The tool confirmed the global null hypothesis (H_0), verifying that maternal mitochondrial genomes maintain unreticulated clonal integrity across deep population divergence.

Multi-Way Mosaicism and Unsourced Lineage Introgression

Natural viral recombination frequently involves more than two parental lineages. Triplets cannot resolve swarms. When circulating recombinant forms undergo secondary cross-over events with tertiary strains, the resulting mosaic genomes contain interleaved tracts from three or more parental sources. Triplet methods face an intrinsic mathematical ceiling: they examine $\binom{N}{3}$ subgraphs in isolation, assuming each recombinant child descends from exactly two parents (19). To test multi-parent resolution, we evaluated a multi-way mosaic recombinant synthesized from three distinct parental lineages across four alternating breakpoints (Trial 88015: $[P1 \mid 1-182] \rightarrow [P2 \mid 183-412] \rightarrow [P1 \mid 413-564] \rightarrow [P3 \mid 565-762] \rightarrow [P1 \mid 763-1,200]$, $N = 64$ taxa).

The triplet approach suffered immediate structural failure. Because 3SEQ evaluates isolated triplets (P, Q, C), no single triplet contains all three parental sequences simultaneously. When testing candidate triplet ($P1, P3, R$), 3SEQ localized the transition between $P1$ and $P3$ at nucleotides 562.0 and 758.5 (mean absolute error 2.8 nt), but the intervening presence of $P2$ sequence corrupted the prefix match score. 3SEQ completely missed the first donor cassette ([182, 412]) and failed to detect two of the four true breakpoints. Structural blindness truncates multi-way chimeras. In contrast, **RhizAeon** operates over the full $N \times U$ attention tensor without triplet downsampling. Dynamic velocity reversal deconvolved both orthogonal spatial modes, detecting the $P2$ cassette at nucleotides 182–412 and the $P3$ cassette at nucleotides 564–762. **RhizAeon** resolved all four breakpoints in 44.9 milliseconds, achieving a 133-fold speedup over 3SEQ (5,971.7 ms).

A second structural challenge arises when one parental lineage is unsourced or extinct. Ghosts confuse symmetric tests. In Trial 76010, recombinant R acquired a 1,165-nucleotide cassette from an unsourced ghost donor ($N = 32$ taxa, $L = 3,000$ nt, $\bar{d} = 0.123$). Because the true donor was absent from the reference panel, 3SEQ's symmetric candidate comparison broke down, emitting 33 distinct statistically significant recombinant triplets ($p < 10^{-10}$) and falsely identifying innocent background clonal leaves ($O, P1, P2 \dots P31$) as recombinant children. Combinatorial methods hallucinate under missing data. **RhizAeon** recognized that trajectory R underwent dynamic velocity reversal without an opposing contemporary pole, isolating the introgression as a single unified reticulation in 29.3 milliseconds without generating false positives among sister taxa.

We corroborated these findings on primary biological isolates using the canonical HIV-1 circulating recombinant form CRF02_AG isolate KAL-153 ($N = 9$ taxa, $L = 9,953$ nt, 1,554 informative sites). KAL-153 represents a complex Subtype A and Subtype G mosaic across the full viral genome (5). **RhizAeon** scanned the complete near-full-length genome in 284.3 milliseconds, resolving two primary reticulation events: a

5'-terminal introgression from Subtype A into Subtype B background spanning nucleotides 1–2,780 (Fisher exact $p = 2.14 \times 10^{-55}$, $Z = 7.15$) and a 3'-terminal cassette spanning nucleotides 8,756–9,638 ($p = 3.82 \times 10^{-14}$, $Z = 10.13$). Whole genomes yield clean breaks.

Computational Scalability and Real-Time Genomic Surveillance

Genomic epidemiology requires algorithms that scale gracefully with cohort size. Sequencing volume swamps cubic tools. Methods that enumerate triplets scale as $O(N^3 \cdot L)$, evaluating $\binom{N}{3} = \frac{N(N-1)(N-2)}{6}$ sequence combinations. For $N = 64$ sequences, 3SEQ tests 41,664 triplets, requiring 5.97 seconds on benchmark alignments. At $N = 128$, the triplet burden expands to 341,376 combinations, requiring 550 seconds. For routine surveillance cohorts of $N = 1,000$ genomes, exhaustive triplet testing demands approximately 166 million comparisons, ballooning runtimes to multiple days. Cubic complexity halts continuous surveillance.

RhizAeon shatters the cubic bottleneck. By projecting genomes into continuous Buneman metric space against $K \leq 32$ spectral landmarks, trajectory extraction scales strictly as $O(N \cdot K \cdot L)$. On identical benchmark alignments, **RhizAeon** accelerated execution across all cohort sizes. At $N = 16$, **RhizAeon** executed 7.4-fold faster than 3SEQ (12.8 ms vs 95.1 ms). The speedup expanded to 52.4-fold at $N = 32$ (29.3 ms vs 1,537.7 ms), 133-fold at $N = 64$ (44.9 ms vs 5,971.7 ms), and 500-fold at $N = 128$ (1.16 s vs 550 s). Linear scaling enables large cohorts.

We tested real-world high-density surveillance using a 25-genome cohort of Potato Virus Y (PVY, $L = 9,594$ nt, 5,249 informative sites). Potyviruses exhibit intense reticulate evolution in agricultural crops (10). **RhizAeon** scanned the alignment in 1.69 seconds, isolating ten verified recombination events across four distinct parental backgrounds. The tool localized both 5'-terminal crossovers (such as sequence H at nucleotide 2,328) and internal cassettes without manual window tuning. To enable friction-free deployment, **RhizAeon** compiles to native Rust command-line binaries and zero-install client-side WebAssembly. Researchers execute full genomic scans within standard web browsers in seconds, processing sensitive pathogen sequences locally without transmitting private genomic data over external networks.

Discussion and Algorithmic Scaling

Linear Manifold Geometry vs. Combinatorial Triplet Search

Computational phylogenetic methods confront a difficult choice between evolutionary realism and combinatorial intractability. Trees model divergence. When reticulate events occur, forcing mosaic genomes into strictly bifurcating topologies distorts branch lengths and inflates substitution rate estimates (26). Conversely, modeling recombination explicitly through ancestral recombination graphs or phylogenetic changepoints introduces prohibitive computational scaling, demanding cubic or quartic algorithms that collapse on modern genomic datasets (22). Triplet scanning methods like 3SEQ provide an empirical compromise by evaluating small three-sequence subgraphs (2). Triplet decomposition avoids tree estimation. Yet exhaustive triplet testing scales cubically with sample size ($\binom{N}{3} \sim O(N^3)$), creating an inescapable computational barrier when surveillance cohorts expand to hundreds or thousands of isolates.

RhizAeon demonstrates that continuous metric geometry dissolves this dichotomy. By projecting whole-genome sequence divergence into fixed Buneman coordinates once for the entire alignment, the engine establishes an invariant spatial reference frame. Zero Procrustes rotations are required. Instantaneous sequence affinities are tracked continuously along the chromosome via analytical Gower out-of-sample projection, replacing discrete topology searches with smooth dynamical trajectories. Manifolds preserve spatial continuity. Cross-taxa attention operates against a fixed landmark basis ($K \leq 32$), ensuring that trajectory extraction scales strictly as $O(N \cdot K \cdot L)$. As our empirical benchmarks demonstrate, this linear scaling yields two- to three-order-of-magnitude accelerations over triplet tools on standard cohorts, finishing multi-kilobase scans in milliseconds where triplet enumeration requires minutes or hours.

Equally consequential is the resolution of complex multi-way mosaicism. Triplet tools assume two-parent ancestry. When circulating recombinant lineages accumulate segments from three or more parental lineages, examining triplets in isolation inevitably truncates mosaic tracts, missing breakpoints and hallucinating recombinant candidates. By tracking continuous trajectory tensors across the entire cohort simultaneously, physical force field deconvolution decomposes multiple orthogonal spatial modes. The framework detects interleaved parental contributions without prior constraints on parental cardinality. Furthermore, metric isolation prevents unsampled ghost donors from triggering combinatorial false alarms across innocent contemporary clades.

Data Governance and Biosafety Compliance

Research integrity demands strict transparency regarding computational protocols and data sources. Computational analyses require ethical clarity. We certify that this investigation represents a purely retrospective bioinformatic study operating exclusively on publicly accessible nucleotide sequences deposited in NCBI GenBank, GISAID, and the Los Alamos National Laboratory HIV Sequence Database. No biological specimens, clinical samples, or viral cultures were collected, isolated, cultivated, or manipulated during this study. All research was conducted entirely in silico using audited open-source algorithms. The software performs automated sequence alignment scanning without generating actionable protocols for pathogen enhancement, gain-of-function phenotypes, or laboratory synthesis. Source code, documentation, and client-side browser modules are freely available under permissive open-source licensing at <https://github.com/veg/rhizaen-code>, promoting reproducible evolutionary genomics and accessible public health surveillance.

Materials and Methods

Farthest-Point Landmark Sampling and Metric Basis Construction

Let $\mathcal{S} = \{s_1, s_2, \dots, s_M\}$ denote a multiple sequence alignment of M contemporary nucleotide sequences of length L over the alphabet $\Sigma = \{A, C, G, T\} \cup \{-\}$. When $M \gg 1,000$, computing an all-to-all pairwise distance matrix requires $\binom{M}{2} \approx \frac{1}{2}M^2$ pairwise comparisons, scaling as $O(M^2L)$ and creating a computational bottleneck for large-scale genomic surveillance. Diversity anchors coordinates. **RhizAeon** eliminates this quadratic barrier through sequential Farthest-Point Sampling (FPS) (11), selecting a sparse subset of $N \ll M$ reference landmark genomes in strictly $O(NML)$ operations.

The selection algorithm begins by designating the initial reference landmark $s_{(1)}$ as the sequence exhibiting the highest non-gap coverage across the alignment:

$$s_{(1)} = \arg \max_{s_i \in \mathcal{S}} \sum_{u=1}^L \mathbb{I}(s_i(u) \neq '-'). \quad (1)$$

The algorithm then computes the pairwise evolutionary distance from $s_{(1)}$ to all M sequences in the cohort, initializing a running minimum distance vector $\mathbf{r} = (r_1, \dots, r_M)^T \in \mathbb{R}^M$, where $r_i = d(s_i, s_{(1)})$. Pairwise evolutionary distances $d(s_i, s_j)$ are computed under the Tamura-Nei (TN93) continuous-time Markov substitution model with pairwise deletion of gap positions (31):

$$d_{\text{TN93}} = -2a_1 \ln \left(1 - \frac{P_1}{2a_1} - \frac{Q}{2a_3} \right) - 2a_2 \ln \left(1 - \frac{P_2}{2a_2} - \frac{Q}{2a_3} \right) - 2(a_3 - a_1 - a_2) \ln \left(1 - \frac{Q}{2a_3} \right), \quad (2)$$

where P_1 and P_2 represent transitional differences between purines and pyrimidines, Q represents transversional differences, and a_1, a_2, a_3 denote equilibrium nucleotide frequency constants. For alignments of highly divergent or non-coding sequences, normalized Hamming distance is supported as a scale-invariant alternative.

Each subsequent reference landmark $k \in \{2, \dots, N\}$ is selected sequentially by identifying the sequence that maximizes the minimum distance to the existing set of selected landmarks:

$$s_{(k)} = \arg \max_{i \in \{1, \dots, M\} \setminus \{s_{(1)}, \dots, s_{(k-1)}\}} r_i. \quad (3)$$

Upon selecting $s_{(k)}$, the algorithm evaluates distances from $s_{(k)}$ to all M sequences and updates the running minimum distance array in $O(ML)$ time:

$$r_i \leftarrow \min(r_i, d(s_i, s_{(k)})), \quad \forall i \in \{1, \dots, M\}. \quad (4)$$

Selecting N reference genomes across M isolates requires exactly $N \times M$ distance evaluations, reducing the computational burden from $O(M^2L)$ to $O(NML)$. For an alignment of 10,000 genomes with $N = 16$ landmarks, distance evaluations collapse from 49,995,000 to 160,000, representing a 312-fold empirical speedup. Farthest points span divergence.

From the selected reference genomes $\{s_{(1)}, \dots, s_{(N)}\}$, we construct the symmetric reference distance matrix $\mathbf{D} \in \mathbb{R}^{N \times N}$, with elements $D_{i,j} = d(s_{(i)}, s_{(j)})$ and diagonal entries $D_{i,i} = 0$. We define the element-wise squared distance matrix $\mathbf{D}^{(2)} \in \mathbb{R}^{N \times N}$ such that $[\mathbf{D}^{(2)}]_{i,j} = D_{i,j}^2$. To map these distances into an inner-product space, we apply the geometric centering projection operator $\mathbf{H} = \mathbf{I}_N - \frac{1}{N} \mathbf{1}_N \mathbf{1}_N^T$, where $\mathbf{I}_N$ is the $N \times N$ identity matrix and $\mathbf{1}_N = (1, \dots, 1)^T \in \mathbb{R}^N$ is the all-ones vector. The doubly centered inner-product Gram matrix $\mathbf{B} \in \mathbb{R}^{N \times N}$ is formulated via classical multidimensional scaling (MDS) (12):

$$\mathbf{B} = -\frac{1}{2} \mathbf{H} \mathbf{D}^{(2)} \mathbf{H}. \quad (5)$$

Spectral decomposition of the symmetric matrix $\mathbf{B}$ yields:

$$\mathbf{B} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T = \sum_{k=1}^N \lambda_k \mathbf{v}_k \mathbf{v}_k^T, \quad (6)$$

where $\mathbf{\Lambda} = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_N)$ contains real eigenvalues arranged in descending order ($\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_N$), and $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_N] \in \mathbb{R}^{N \times N}$ is the orthonormal eigenvector matrix satisfying $\mathbf{V}^T \mathbf{V} = \mathbf{I}_N$. Spectra reveal splits.

We retain the leading d dimensions corresponding to the largest positive eigenvalues ($d \leq \min(4, N-1)$):

$$\mathbf{\Lambda}_d = \text{diag}(\lambda_1, \dots, \lambda_d) \in \mathbb{R}^{d \times d}, \quad \mathbf{V}_d = [\mathbf{v}_1, \dots, \mathbf{v}_d] \in \mathbb{R}^{N \times d}. \quad (7)$$

The static d -dimensional metric coordinates $\mathbf{x}_i \in \mathbb{R}^d$ for reference landmark $i \in \{1, \dots, N\}$ are given by:

$$\mathbf{x}_i = \mathbf{\Lambda}_d^{1/2} (\mathbf{V}_d)_{i,:}^T = \left(\sqrt{\lambda_1} v_{1,i}, \sqrt{\lambda_2} v_{2,i}, \dots, \sqrt{\lambda_d} v_{d,i} \right)^T. \quad (8)$$

By construction, these coordinates satisfy $\langle \mathbf{x}_i, \mathbf{x}_j \rangle = [\mathbf{B}]_{i,j}$ within the subspace $\mathbb{R}^d$. By Buneman's metric tree theorem (4), tree metrics satisfy four-point additivity conditions $d(a,b) + d(c,d) \leq \max(d(a,c) + d(b,d), d(a,d) + d(b,c))$, ensuring that tree-like evolutionary divergences embed into Euclidean spaces where macroscopic clade splits project onto the dominant eigenvectors.

Analytical Gower Out-of-Sample Trajectory Projection

Tracking sequence affinities along the chromosome requires establishing an instantaneous coordinate $\mathbf{z}_i(u) \in \mathbb{R}^d$ for each query sequence $i \in \{1, \dots, M\}$ at every genomic site $u \in [1, L]$. Recomputing multidimensional scaling independently within sliding windows would produce arbitrary orthogonal rotations and reflections, distorting geometric distances and necessitating expensive Procrustes alignments. Instead, **RhizAeon** projects local distance vectors into the fixed canonical coordinate basis $(\mathbf{V}_d, \mathbf{\Lambda}_d)$ using Gower's analytical out-of-sample interpolation theorem (13).

At coordinate u , we evaluate local evolutionary distances across an aligned window $[u - W/2, u + W/2]$ of width W (default $W = 300$ nt) sampled at stride S (default $S = 10$ nt). For query sequence i , this yields an instantaneous local distance vector $\mathbf{d}_i(u) \in \mathbb{R}^N$ to the N reference landmarks:

$$\mathbf{d}_i(u) = (d_i(1; u), d_i(2; u), \dots, d_i(N; u))^T. \quad (9)$$

We define the vector of squared local distances as $\mathbf{d}_i^2(u) = (d_i(1; u)^2, \dots, d_i(N; u)^2)^T \in \mathbb{R}^N$, and the vector of mean squared reference distances as:

$$\bar{\mathbf{d}}^2 = \frac{1}{N} \mathbf{D}^{(2)} \mathbf{1}_N = \left(\frac{1}{N} \sum_{j=1}^N D_{1,j}^2, \dots, \frac{1}{N} \sum_{j=1}^N D_{N,j}^2 \right)^T \in \mathbb{R}^N. \quad (10)$$

Gower's out-of-sample mapping identifies the coordinate $\mathbf{z}_i(u) \in \mathbb{R}^d$ that minimizes squared Euclidean distance distortions relative to the static reference coordinates $\mathbf{x}_j$:

$$\min_{\mathbf{z} \in \mathbb{R}^d} \sum_{j=1}^N (\|\mathbf{z} - \mathbf{x}_j\|^2 - d_i(j; u)^2)^2. \quad (11)$$

Expanding the Euclidean norm $\|\mathbf{z} - \mathbf{x}_j\|^2 = \|\mathbf{z}\|^2 - 2\mathbf{z}^T \mathbf{x}_j + \|\mathbf{x}_j\|^2$, utilizing the centering condition $\sum_{j=1}^N \mathbf{x}_j = \mathbf{0}$, and equating the gradient to zero yields the exact analytical solution:

$$\mathbf{z}_i(u) = \frac{1}{2} \mathbf{\Lambda}_d^{-1/2} \mathbf{V}_d^T (\bar{\mathbf{d}}^2 - \mathbf{d}_i^2(u)) \in \mathbb{R}^d, \quad (12)$$

where $\mathbf{\Lambda}_d^{-1/2} = \text{diag}(1/\sqrt{\lambda_1}, \dots, 1/\sqrt{\lambda_d})$. Evaluating (12) across all coordinates $u \in \{1, 1 + S, \dots, L\}$ constructs the continuous trajectory tensor $\mathbf{Z} \in \mathbb{R}^{M \times U \times d}$, where $U = \lceil L/S \rceil$.

Because the basis $(\mathbf{V}_d, \mathbf{\Lambda}_d)$ and mean vector $\bar{\mathbf{d}}^2$ are computed once from whole-genome reference divergence, every sliding window maps into the identical, rigid metric reference frame. Zero Procrustes rotations are required. Projections preserve geometry. Furthermore, substituting the whole-genome distance vector of reference landmark k into (12) analytically recovers its exact MDS coordinate:

$$\frac{1}{2} \mathbf{\Lambda}_d^{-1/2} \mathbf{V}_d^T \left(\bar{\mathbf{d}}^2 - (\mathbf{D}_{k,:}^{(2)})^T \right) = \mathbf{\Lambda}_d^{1/2} (\mathbf{V}_d)_k^T = \mathbf{x}_k, \quad (13)$$

proving algebraic self-consistency and zero metric distortion on reference taxa.

Generalized Tree-RoPE and Continuous-Time Markov Substitution Prior

Standard transformer neural networks incorporate sequential position along one-dimensional text using Rotary Position Embeddings (RoPE) (29). However, biological sequences diverge across multidimensional phylogenetic trees rather than linear strings. To integrate evolutionary distance directly into cross-sequence attention, **RhizAeon** generalizes rotary embeddings from one-dimensional sequence indices to continuous metric coordinates.

Let $\mathbf{x}_i \in \mathbb{R}^d$ denote the continuous metric coordinates of sequence i . We construct a fixed frequency projection basis $\mathbf{\Theta} \in \mathbb{R}^{(d_{\text{head}}/2) \times d}$, with components parameterized by base frequency $\theta_0 = 10,000$:

$$\mathbf{\Theta}_{m,c} = \theta_0^{-2(m-1)/d_{\text{head}}} \cdot \omega_c, \quad m \in \{1, \dots, d_{\text{head}}/2\}, c \in \{1, \dots, d\}, \quad (14)$$

where $\omega_c = \sqrt{\lambda_c / \sum_{k=1}^d \lambda_k}$ weights each dimension by its relative spectral variance. For sequence i , the rotary phase angle vector $\boldsymbol{\theta}_i = (\theta_{i,1}, \dots, \theta_{i,d_{\text{head}}/2})^T$ is computed as:

$$\theta_{i,m} = \sum_{c=1}^d x_{i,c} \mathbf{\Theta}_{m,c}. \quad (15)$$

Angles encode divergence.

At each aligned site u , sequence representations generate query and key vectors $\mathbf{q}_i(u), \mathbf{k}_i(u) \in \mathbb{R}^{d_{\text{head}}}$. We define the block-diagonal Givens rotation matrix $\mathbf{R}(\boldsymbol{\theta}_i) \in \mathbb{R}^{d_{\text{head}} \times d_{\text{head}}}$ as:

$$\mathbf{R}(\boldsymbol{\theta}_i) = \bigoplus_{m=1}^{d_{\text{head}}/2} \begin{pmatrix} \cos \theta_{i,m} & -\sin \theta_{i,m} \\ \sin \theta_{i,m} & \cos \theta_{i,m} \end{pmatrix}. \quad (16)$$

Applying $\mathbf{R}(\boldsymbol{\theta}_i)$ rotates query and key representations according to each taxon's continuous metric position. The inner product between rotated vectors satisfies:

$$\langle \mathbf{R}(\boldsymbol{\theta}_i) \mathbf{q}_i(u), \mathbf{R}(\boldsymbol{\theta}_j) \mathbf{k}_j(u) \rangle = \mathbf{q}_i(u)^T \mathbf{R}(\boldsymbol{\theta}_j - \boldsymbol{\theta}_i) \mathbf{k}_j(u). \quad (17)$$

Because rotation angles depend linearly on metric coordinates, the phase shift $\|\boldsymbol{\theta}_j - \boldsymbol{\theta}_i\|$ increases with evolutionary divergence in Buneman metric space, naturally attenuating pairwise attention as phylogenetic distance increases.

To calibrate geometric distances against baseline neutral sequence homology, we integrate a continuous-time Markov substitution prior directly into the attention logits. For any pair of sequences (i, j) separated by evolutionary distance $D_{i,j}$, the probability of observing identical homologous nucleotides under a continuous-time Markov substitution model with substitution scaling rate λ_h and equilibrium frequency $\epsilon_0 = 0.25$ is:

$$P_{i,j} = \epsilon_0 + (1 - \epsilon_0) \exp(-\lambda_h D_{i,j}). \quad (18)$$

We define the logarithmic phylogenetic prior bias as:

$$\text{PhyloBias}_{i,j} = \ln [\max(10^{-5}, P_{i,j})]. \quad (19)$$

The composite unnormalized cross-attention score $S_u(i, j)$ between query sequence i and candidate reference j at coordinate u becomes:

$$S_u(i, j) = \frac{(\mathbf{R}(\boldsymbol{\theta}_i) \mathbf{q}_i(u))^T (\mathbf{R}(\boldsymbol{\theta}_j) \mathbf{k}_j(u))}{\sqrt{d_{\text{head}}}} + \text{PhyloBias}_{i,j}. \quad (20)$$

Unpolarized Root Sink Architecture and Autapomorphy Absorption

Standard cross-attention mechanisms normalize probability exclusively across contemporary leaves, creating structural vulnerability to lineage-specific rate variation. Private mutations lack donors. When an isolated lineage undergoes an episodic substitution burst—such as APOBEC3G cytidine deamination or accelerated heterotachy—private mutations lack homologous counterparts in contemporary clades. Without an unpolarized reference, softmax forces these private mutations to affiliate with whichever contemporary leaf presents the smallest stochastic mismatch, generating severe false positive recombination calls.

To insulate trajectory inference from unreticulated rate bursts, **RhizAeon** reserves index zero in the sequence dimension for an unpolarized ancestral root token $\mathbf{x}_{\text{root}}$. The root token is anchored at the origin of the Buneman metric space:

$$\mathbf{x}_{\text{root}} = \mathbf{0} = (0, 0, \dots, 0)^T \in \mathbb{R}^d. \quad (21)$$

For an alignment of N contemporary reference genomes, the cross-attention operator operates over $N + 1$ nodes, spanning the ancestral root ($j = 0$) and all contemporary leaves ($j = 1, \dots, N$). The softmax normalization strictly conserves probability across all $N + 1$ nodes:

$$A_u(i \rightarrow j) = \frac{\exp(S_u(i, j))}{\sum_{k=0}^N \exp(S_u(i, k))}, \quad \text{such that} \quad A_u(i \rightarrow 0) + \sum_{j=1}^N A_u(i \rightarrow j) = 1.0. \quad (22)$$

Proof of Autapomorphy Absorption Consider a genomic site u where sequence i acquires a private autapomorphic mutation that does not match any contemporary reference genome $j \in \{1, \dots, N\}$. For all contemporary leaves $j \geq 1$, the mismatch depresses both query-key matching and the phylogenetic prior, yielding low scores $S_u(i, j) \leq S_{\text{mismatch}}$. In contrast, the unpolarized root token at $\mathbf{x}_{\text{root}} = \mathbf{0}$ represents the baseline unmutated ancestral state with neutral substitution distance, maintaining $S_u(i, 0) = S_{\text{root}} > S_{\text{mismatch}}$. In the limit where private divergence increases:

$$\lim_{S_u(i, j) \rightarrow -\infty, \forall j \geq 1} A_u(i \rightarrow 0) = \frac{\exp(S_u(i, 0))}{\exp(S_u(i, 0)) + \sum_{j=1}^N \exp(S_u(i, j))} = 1.0, \quad (23)$$

and $A_u(i \rightarrow j) \rightarrow 0.0$ for all $j \geq 1$. Consequently, autapomorphic substitution bursts route their attention flux into node zero rather than contemporary leaves, preventing spurious donor excursions and guaranteeing a 0.00% false positive rate under biological heterotachy. $\square$

Sinks absorb outliers.

Continuous Trajectory Integration and Brownian Bridge Inference

At individual polymorphic sites, instantaneous attention weights exhibit stochastic high-frequency jitter. To extract continuous evolutionary trajectories, **RhizAeon** integrates attention into cumulative trajectory curves. Let $\mathcal{B}_m \subset \{1, \dots, N\}$ denote the set of reference genomes belonging to phylogenetic clade m . The instantaneous bundle attention is $\alpha_{i,m}(u) = \sum_{j \in \mathcal{B}_m} A_u(i \rightarrow j)$. The continuous cumulative attention curve $C_{i,m}(u)$ represents the integrated affinity to clade m up to coordinate u :

$$C_{i,m}(u) = \sum_{v=1}^u \alpha_{i,m}(v) = \sum_{v=1}^u \sum_{j \in \mathcal{B}_m} A_v(i \rightarrow j). \quad (24)$$

The home clade $m_{\text{home}}(i)$ is identified as the lineage accumulating the greatest integrated affinity across the full chromosome:

$$m_{\text{home}}(i) = \arg \max_m C_{i,m}(L). \quad (25)$$

For any candidate donor clade $m \neq m_{\text{home}}$, we formulate the differential attention trajectory as:

$$Y_{i,m}(u) = C_{i,m}(u) - C_{i,m_{\text{home}}}(u). \quad (26)$$

Under clonal evolution within the parental background, attention to the home clade exceeds donor attention at nearly all sites ($A_v(i \rightarrow \text{home}) > A_v(i \rightarrow m)$), causing $Y_{i,m}(u)$ to drift monotonically downward with negative slope. Recombination reverses the derivative: within a converted donor tract $[u_1, u_2]$, donor attention dominates, producing a positive slope ($dY/du > 0$) and a prominent upward excursion. Bridges bound excursions.

To test the clonal null hypothesis H_0 , we construct the centered Brownian bridge process $B_{i,m}(u)$ by subtracting the chromosome-wide linear drift:

$$B_{i,m}(u) = Y_{i,m}(u) - \frac{u}{L} Y_{i,m}(L), \quad u \in [1, L]. \quad (27)$$

Let $\Delta Y(u) = Y(u) - Y(u-1)$ denote the local trajectory increments, with sample mean $\bar{\Delta Y} = \frac{1}{L} Y(L)$ and empirical increment variance:

$$\sigma^2 = \frac{1}{L-1} \sum_{u=1}^L (\Delta Y(u) - \bar{\Delta Y})^2. \quad (28)$$

Under the clonal null hypothesis H_0 , the increments are stationary and unreticulated. By Donsker's functional central limit theorem (9), the standardized process:

$$W_L^0(t) = \frac{B_{i,m}(\lfloor tL \rfloor)}{\sigma\sqrt{L}}, \quad t \in [0, 1], \quad (29)$$

converges weakly as $L \rightarrow \infty$ to a standard Brownian bridge $W^0(t)$ with continuous sample paths, mean $\mathbb{E}[W^0(t)] = 0$, and covariance $\text{Cov}(W^0(s), W^0(t)) = \min(s, t) - st$. The supremum of the absolute bridge trajectory follows the two-sided Kolmogorov distribution:

$$P\left(\sup_{t \in [0,1]} |W^0(t)| \leq z\right) = 1 - 2 \sum_{k=1}^{\infty} (-1)^{k-1} \exp(-2k^2 z^2). \quad (30)$$

To control the family-wise error rate (FWER) across all $M_{\text{topo}} = \binom{N}{2} \max(1, N-2)$ testable donor-recipient reticulation topologies at nominal significance $\alpha = 0.05$, we apply the Dunn-Sidak extreme-value correction (27):

$$K_{\text{crit}} = \sqrt{-\frac{1}{2} \ln(1 - (1 - \alpha)^{1/M_{\text{topo}}})} \approx \sqrt{-\frac{1}{2} \ln\left(\frac{\alpha}{2M_{\text{topo}}}\right)}. \quad (31)$$

If the normalized trajectory supremum satisfies:

$$\frac{\sup_{u \in [1, L]} |B_{i,m}(u)|}{\sigma\sqrt{L}} \geq K_{\text{crit}}, \quad (32)$$

RhizAeon rejects the clonal null hypothesis and confirms a statistically significant reticulation event.

Dyadic Scale-Space Matched Filtering and Running Drawdown

For short genetic conversions and micro-recombinations (< 200 nt), global Brownian bridge tests can suffer signal dilution across multi-kilobase chromosomes. To achieve multi-scale sensitivity, **RhizAeon** convolves the differential trajectory flux $\Delta Y(u)$ with a bank of zero-mean bipolar dyadic matched filters $h_\tau(s)$ evaluated across scales $\tau \in \{15, 25, 50, 100, 200, 500\}$ nucleotides:

$$h_\tau(s) = \begin{cases} +1, & 0 \leq s < \tau/2 \\ -1, & \tau/2 \leq s < \tau \\ 0, & \text{otherwise} \end{cases} \quad (33)$$

The discrete convolution $(dY * h_\tau)(u) = \sum_{s=0}^{\tau} \Delta Y(u-s) h_\tau(s)$ isolates rapid step transitions across dyadic scale space. Wavelets isolate tracts.

Concurrently, **RhizAeon** tracks the running high-water mark $M(u)$ and running drawdown $DD(u)$:

$$M(u) = \max_{1 \leq v \leq u} Y(v), \quad DD(u) = M(u) - Y(u). \quad (34)$$

The drawdown metric localizes points where trajectory momentum collapses, marking the exit boundary u_2 of a donor tract. To distinguish single crossovers from intercalary cassette swaps, we evaluate the normalized energy density ratio of the trajectory excursions outside versus inside the candidate tract $[u_1, u_2]$:

$$\mathcal{E}_{\text{ratio}} = \frac{\frac{1}{L - (u_2 - u_1)} \int_{u \notin [u_1, u_2]} (Y(u) - Y_{\text{base}})^2 du}{\frac{1}{u_2 - u_1} \int_{u_1}^{u_2} (Y(u) - Y_{\text{base}})^2 du}, \quad (35)$$

where Y_{base} denotes the linear baseline drift. A single crossover spans half the chromosome, producing high residual energy across the alignment ($\mathcal{E}_{\text{ratio}} \geq 0.50$). In contrast, an intercalary cassette returns cleanly to baseline on both flanks, characterized by $\mathcal{E}_{\text{ratio}} < 0.15$.

Physical Force Field Deconvolution and Random Matrix Halting

To deconvolve complex multi-way mosaicism, **RhizAeon** models trajectory excursions as physical displacements within an evolutionary force field. We parameterize the evolutionary pull of the home lineage as a Hookean restoring potential $U(\mathbf{z}) = \frac{1}{2} \lambda \|\mathbf{z} - \mathbf{z}_{\text{home}}\|^2$, generating a continuous restoring force:

$$\mathbf{f}(u) = -\nabla U = -\lambda(\mathbf{z}(u) - \mathbf{z}_{\text{home}}) \in \mathbb{R}^d. \quad (36)$$

Across a candidate breakpoint boundary, let $\mathbf{f}_{\text{left}}$ and $\mathbf{f}_{\text{right}}$ denote the mean force vectors on the flanking intervals. The directional deflection angle θ satisfies $\cos \theta = \frac{\langle \mathbf{f}_{\text{left}}, \mathbf{f}_{\text{right}} \rangle}{\|\mathbf{f}_{\text{left}}\| \|\mathbf{f}_{\text{right}}\|}$, and the step velocity is $v_{\text{step}} = \|\mathbf{f}_{\text{right}} - \mathbf{f}_{\text{left}}\|$. The dynamic velocity reversal score is defined as:

$$R = v_{\text{step}}(1 - \cos \theta). \quad (37)$$

Forces deconvolve modes. To achieve scale-invariance across cohorts of varying divergence, we standardize the velocity reversal along the unit parental dipole axis $\mathbf{d}_{\text{hat}} = \frac{\mathbf{z}_{\text{donor}} - \mathbf{z}_{\text{home}}}{\|\mathbf{z}_{\text{donor}} - \mathbf{z}_{\text{home}}\|}$:

$$Z_{\text{flip}} = \frac{|(\mathbf{f}_{\text{right}} - \mathbf{f}_{\text{left}}) \cdot \mathbf{d}_{\text{hat}}|}{\sqrt{\frac{\sigma_{\text{left}}^2}{L_{\text{left}}} + \frac{\sigma_{\text{right}}^2}{L_{\text{right}}}}}, \quad (38)$$

where $\sigma_{\text{left}}^2, \sigma_{\text{right}}^2$ represent residual force variances along $\mathbf{d}_{\text{hat}}$ across flanking segments of lengths $L_{\text{left}}, L_{\text{right}}$.

For alignments exhibiting multiple recombinant tracts, we stack force vectors into the spatio-temporal force matrix $\mathbf{F} \in \mathbb{R}^{N \times L}$ and perform singular value decomposition:

$$\mathbf{F} = \sum_{m=1}^r s_m \mathbf{u}_m \mathbf{v}_m^T, \quad (39)$$

where $s_1 \geq s_2 \geq \dots \geq s_r$ denote singular values. Each detected reticulation mode m is deconvolved via iterative mode peeling, subtracting $s_m \mathbf{u}_m \mathbf{v}_m^T$ from $\mathbf{F}$. To prevent over-segmentation and halt peeling when residual variation reflects neutral coalescent drift, we apply random matrix theory. Under the unreticulated null hypothesis, the singular values of an $N \times L$ random noise matrix follow the Marchenko-Pastur distribution (18). The theoretical upper boundary of the bulk spectral noise edge is:

$$s_{\text{bulk}} = \sigma \sqrt{L} \left(1 + \sqrt{\frac{N}{L}} \right), \quad (40)$$

where σ represents the baseline standard deviation of empirical force increments. Mode peeling halts when the leading residual singular value satisfies $s_m \leq s_{\text{bulk}}$, terminating iteration without manual parameter tuning.

Profile Maximum Likelihood Breakpoint Polishing

Following changepoint localization from continuous trajectory dynamics, **RhizAeon** refines candidate breakpoints down to single-nucleotide resolution using profile maximum likelihood polishing. Within a local search radius $[u_{\text{init}} - \delta, u_{\text{init}} + \delta]$ (default $\delta = 50$ nt), the engine scans each candidate coordinate c and computes the partitioned log-likelihood:

$$\ln \mathcal{L}(c) = \sum_{u=1}^c \ln P(s_{\text{rec}}(u) | s_{\text{left}}(u)) + \sum_{u=c+1}^L \ln P(s_{\text{rec}}(u) | s_{\text{right}}(u)), \quad (41)$$

where $P(s_{\text{rec}}(u) | s(u))$ denotes the conditional probability of the recombinant state given the parental sequence under the fitted nucleotide substitution model.

When invariant or uninformative sites intervene between flanking parental polymorphisms, the likelihood function forms a flat plateau. Rather than reporting an arbitrary midpoint, **RhizAeon** reports the exact informational plateau $[c_{\text{left}}, c_{\text{right}}]$:

$$[c_{\text{left}}, c_{\text{right}}] = \left\{ c \in [1, L] : \ln \mathcal{L}(c) \geq \max_{c'} \ln \mathcal{L}(c') - \Delta_{\text{thresh}} \right\}, \quad (42)$$

where $\Delta_{\text{thresh}} = 1.92$ corresponds to the 95% confidence interval under a χ^2_1 likelihood ratio test. The boundary coordinates c_{left} and c_{right} demarcate the exact flanking informative polymorphisms that physically bound the genetic exchange. Plateaus bound breakpoints.

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